Shahenshah Ali Syed

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Profile

- **Software Developer (Data/AI)** with over **15 years of experience** delivering enterprise-grade data engineering solutions across **Healthcare, Retail, Banking, Finance, and Compliance** industries.
- Specialized in **cloud-native architecture** , **metadata-driven lakehouses**, and **AI-accelerated development**, with a strong record of transforming business outcomes through innovative data strategies.
- Automated infrastructure provisioning and management for cloud-native deployments and orchestrated and scaled containerized applications using docker in production environments.
- Expert in **Google Cloud Platform** (BigQuery, VertexAI, Model garden, DataProc, Composer), **Azure** (Spark,Databricks, Delta Lake), and **AWS**.
- Led the migration of **10PB+ of enterprise data** and built automated lakehouse frameworks, reducing data transformation time by **70%**. Integrated **GitHub Copilot** and **Claude** into the engineering workflow, boosting **code delivery efficiency by 6X** and accelerating release cycles.
- Led API-driven migration of applications and data from on-premises to cloud platforms, ensuring scalability, security, and seamless integration.
- Active hugging face members for open source AI Initiatives.
- Adept at leading agile teams, optimizing pipelines, and aligning technical delivery with business-critical KPIs.

Skills

Cloud Platform — *GCP(BigQuery, VertexAI,GCS,Composer-Airflow 3, Data Proc),Azure (Data factory, Storage,Databricks)* | **Database** — *MySQL,PostgresSQL,MongoDB, Oracle, Teradata* | **Language** — *Python,PySpark, Java,Shell scripting* | **Agile** — *Jira,Confluence* | **DevOps and Container** — *Docker,Terraform, Jenkins, Git cloud, SVN, Splunk* | **Data Analytics AI** — Google ADK,RAG,Hugging Face (NLP) , scikit-learn,TensorFlow | **Intelligence and ETL** — *Actimize Moduler,RCM, Informatica*

Professional Experience

CVS Health, Software Developer (Sr Data Engineer) Nov 2022 – Present | Atlanta, United States

- Leading 4 different initiatives to enable AI use case for Trade and Rebate team.
- Designing and leading any initiatives across the team and working on providing reusable ,scaled solutions.
- Built a semantic search Agent (RAG) to help patients provide medicine-level insights, integrating BigQuery as vector embedding to index endpoints in Vertex AI.

- Designed a reporting framework for Actual vs Forecast profit analysis that saved $120m in the Pharmacy Benefit Management (PBM) segment, using Spark, Azure, and Databricks to identify claim-level and profit variances over fiscal years.
- Defined strategy and implemented a scalable metadata-driven data lakehouse framework to automate ingestion, ETL, data quality checks, auditing, and secure data handling (encryption, decryption, masking), reducing manual effort and achieving 70% time savings.
- Led migration of 1,100+ tables and artifacts from Azure Delta Lake, Teradata, SharePoint, Unix, and AWS Pega to GCP BigQuery, ensuring data consistency and modernizing analytics infrastructure using API's and microservices-based architectures.
- Refactored SQL, legacy code (SAS) into BigQuery stored procedures, improving query performance by 40% and accelerating analytics on large datasets.
- Developed and orchestrated 900+ data pipelines using Airflow(Cloud Composer), serving 6 enterprise tenant teams responsible for forecasting, rebate management, claims tagging, and guaranteed contract tracking.
- Tuned pipeline performance to boost file ingestion speed by 90%, enabling near real-time data availability for operational teams.

Tata Consultancy Services, IT Analyst Aug 2019 – Oct 2022 | Atlanta

- Designed scalable distributed software architecture on GCP to support interactive analytics across large-scale data lake and warehouse environments.
- Migrated ~1PB/year of enterprise data from Hadoop, Hive, DB2, and Teradata into GCP using BigQuery, Dataflow, DataProc, Cloud Storage, Cloud Functions, and Pub/Sub, building end-to-end ETL/ELT solutions.
- Developed and orchestrated 300+ robust data pipelines using Airflow, Cloud Composer, and Jenkins; restructured data dictionaries to streamline processing logic and increase consistency.
- Optimized BigQuery SQL queries for complex business logic, achieving a 45% improvement in query performance and enabling faster turnaround for reporting and insights.
- Engineered dimensional models with Atscale, Tableau, and SSAS, driving enterprise BI initiatives and supporting performance dashboard development across departments.

Tata Consultancy Services, System Engineer Feb 2015 – Jul 2019 | Chennai, India

- Built and maintained Actimize AML screening pipelines to enable 24/7 monitoring of 5,000+ daily transactions totaling ~$10M across global financial networks, ensuring real-time compliance with international block lists.
- Integrated Java-based applications with multiple databases and processed structured messages (SWIFT, XML, MQSeries) to enhance screening accuracy and system performance.
- Designed interactive dashboards using Risk Case Manager to visualize AML alerts including consumer fraud, terrorist financing, global sanctions, and suspicious international money transfers.
- Developed workflows to streamline analyst investigations and facilitate direct reporting of high-risk cases to FINCEN.
- Automated AML operations using custom scripts and web services, boosting system-wide reliability and processing efficiency in fraud detection and regulatory compliance.

Infosys Limited, Associate Business Analyst Oct 2013 – Jan 2015 | Chennai, India

- Designed and developed trade surveillance and fraud analytical models for risk management using Actimize Intelligence, aligning with SEC regulations for equity market compliance.

- Built ETL pipelines for classification models to capture and monitor trading behaviors, ensuring adherence to rules around trade transparency, market abuse, and insider activity.
- Implemented Quantitative compliance workflows for high-risk trading behaviors such as Marking the Close, Insider Trading, Wash Sales, Excessive Order Cancellation, Best Execution, and Low-Priced Securities.
- Collaborated with compliance officers for testing and risk teams to automate case creation and support audit tracking, improving fraud detection accuracy and regulatory response time.

Infosys, Software Developer Sep 2011 – Sep 2013 | Chennai, India

- **Designed and configured a web portal** for managing customer accounts and card transactions, including sell, buy, reload, and cash-out functionalities using Java, Java Servlets, JSP, and DB2.
- **Developed web applications** to facilitate customer account management and various card transaction options, including sell, buy, reload, and cash-out for related products.
- **Implemented web services** using Remote Procedure Call technology and REST APIs to enable third-party transactions, utilizing WSDL provided through the web portal.

Certificates

- Google Cloud Certified–Professional Data Engineer
- Google Cloud Certified - Professional Machine Learning Engineer
- Google Cloud Certified - Generative AI Leader
- Astronomer certified - Apache Airflow 3

Awards

Star Performer, CVS Caremark
Played a significant role in multiple initiatives.

On the Spot Award
Being resilient to tackle the huge outage in Moneygram.

Education

B.Tech In Information Technology Mar 2007 – Apr 2011 | Kolkata, India
West Bengal University And Technology